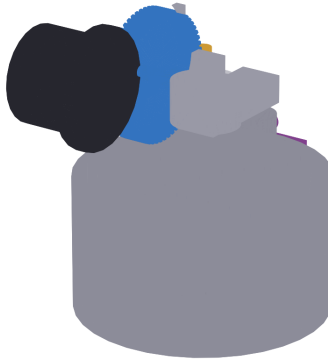


OPTION E — one-third-size pan/tilt mount

A plain-language build guide for a shrunk-down copy of ASSEMBLY_pan_tilt.STEP, redesigned around a real 5V 28BYJ-48 stepper motor. Independent of the parallel "Option D" work on the same source design.



Reference model	ASSEMBLY_pan_tilt.STEP (16 parts / 24 pieces, 245x298x270mm)
This design	22 pieces from 15 real part types (2 non-functional duplicates dropped)
Overall size	about 95 x 74 x 96mm (W x D x H) — see sizing note, Ch. 3
Motor	2x 28BYJ-48, 5V, via ULN2003 driver
Gear ratio	3:1 on both the pan axis and the tilt axis (20-tooth : 60-tooth)
Material	3D-printed plastic body, real off-the-shelf bearings/motor/fasteners
Surprise stat	3 separate rotation axes — pan, tilt, and a free third pivot with no motor

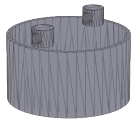
Chapter 1 — What's in the box

Every real, distinct part in this design, one row each. "3D Printed" pieces you make yourself; "Bought" pieces are real off-the-shelf hardware sized to fit.

Part	Source	Qty	What it does
Main housing (base 2)	3D Printed	1	The outer body. Holds the pan bearing at the bottom and carries the two pivot towers on top.
Pan bearing, flanged	Bought	1	Small flanged ball bearing (~5mm bore) the whole upper assembly spins on.
Pan shaft (main shaft)	Bought	1	5mm steel rod running up through the housing; the pan gear and Gear 2 Base key onto it.
Pan motor bracket (Motor base)	3D Printed	1	Holds the pan motor. Its two screw holes are shifted from centre to match the real motor's off-centre shaft.
Motor body	Bought	2	The 28BYJ-48 stepper motor itself — one for pan, one for tilt.
Motor shaft	Bought	2	The motor's own D-shaped output shaft — one for pan, one for tilt.
Small gear, 20 teeth (spur gear PT2)	3D Printed	2	The pinion on each motor's shaft, driving its matching big gear.
Spacer	3D Printed	4	Simple printed tubes that keep gears and bearings the right distance apart on a shaft.
Tilt motor bracket (Gear 2 Base)	3D Printed	1	Keyed onto the top of the pan shaft, so it swings with pan. Carries the tilt motor and the tilt shaft's two bearings.
Tilt shaft, outer half (Shaft 2)	Bought	1	5mm rod the tilt gear rides on, supported by two bearings.
Big gear, 60 teeth, wide (spur gear PT11)	3D Printed	1	The tilt-axis reduction gear — a genuinely separate part from the pan gear, not a duplicate.
Tilt shaft, inner hub (Baring 2 shaft)	3D Printed	1	A short keying hub connecting Shaft 2 to the tilt gear.
Tilt bearing (Wall Ball Bearing)	Bought	2	Small ball bearings supporting the tilt shaft.
Big gear, 60 teeth, narrow (spur gear PT1)	3D Printed	1	The pan-axis reduction gear, keyed to the main shaft.
Cam Holder	3D Printed	1	The largest single part. Carries the free, hand-positioned third pivot and the new laser mount.

Total: 15 real part types, 22 pieces. The original file's 16th and 17th "parts" were two small motor-shaft bearings and a duplicate-looking gear save that turned out on closer inspection to actually be needed — see the note at the end of this chapter.

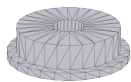
Part-by-part reference



Main housing (base 2)

3D Printed · qty 1

The outer body. Holds the pan bearing at the bottom and carries the two pivot towers on top.



Pan bearing, flanged

Bought · qty 1

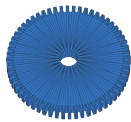
Small flanged ball bearing (~5mm bore) the whole upper assembly spins on.



Pan shaft (main shaft)

Bought · qty 1

5mm steel rod running up through the housing; the pan gear and Gear 2 Base key onto it.



Big gear, 60 teeth, narrow (spur gear PT1)

3D Printed · qty 1

The pan-axis reduction gear, keyed to the main shaft.



Motor body

Bought · qty 2

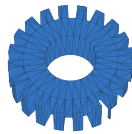
The 28BYJ-48 stepper motor itself — one for pan, one for tilt.



Motor shaft

Bought · qty 2

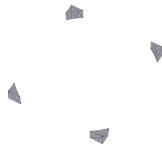
The motor's own D-shaped output shaft — one for pan, one for tilt.



Small gear, 20 teeth (spur gear PT2)

3D Printed · qty 2

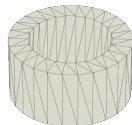
The pinion on each motor's shaft, driving its matching big gear.



Pan motor bracket (Motor base)

3D Printed · qty 1

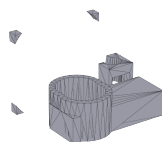
Holds the pan motor. Its two screw holes are shifted from centre to match the real motor's off-centre shaft.



Spacer

3D Printed · qty 4

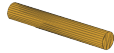
Simple printed tubes that keep gears and bearings the right distance apart on a shaft.



Tilt motor bracket (Gear 2 Base)

3D Printed · qty 1

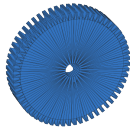
Keyed onto the top of the pan shaft, so it swings with pan. Carries the tilt motor and the tilt shaft's two bearings.



Tilt shaft, outer half (Shaft 2)

Bought · qty 1

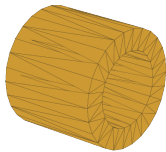
5mm rod the tilt gear rides on, supported by two bearings.



Big gear, 60 teeth, wide (spur gear PT11)

3D Printed · qty 1

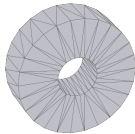
The tilt-axis reduction gear — a genuinely separate part from the pan gear, not a duplicate.



Tilt shaft, inner hub (Baring 2 shaft)

3D Printed · qty 1

A short keying hub connecting Shaft 2 to the tilt gear.



Tilt bearing (Wall Ball Bearing)

Bought · qty 2

Small ball bearings supporting the tilt shaft.



Cam Holder

3D Printed · qty 1

The largest single part. Carries the free, hand-positioned third pivot and the new laser mount.

Chapter 2 — How it moves

This design has three separate places where something rotates. Two are driven by motors; one is free-swinging and positioned by hand.

Axis	Driven by	How it moves
Pan (spin left/right)	Pan motor + 3:1 gear pair	The pan motor turns a small 20-tooth gear, which turns the big 60-tooth gear keyed to the main shaft. Everything above the main housing — Gear 2 Base, the tilt motor, and eventually the Cam Holder assembly — spins together with it.
Tilt (nod up/down)	Tilt motor + 3:1 gear pair	The same arrangement, mounted on Gear 2 Base instead of the main housing, so it travels around with the pan motion. Turns Shaft 2 through its own 60-tooth gear.
Third pivot (Cam Holder)	Nothing — free, by hand	The Cam Holder's own pivot pins sit loosely in two round sockets on the main housing's twin towers. This is not connected to either motor gear train — it was already like this in the original reference file, and per your instruction we kept it exactly that way rather than motorising or locking it.

Both motor-driven axes use the same reduction: a 20-tooth gear on the motor's shaft turns a 60-tooth gear, a 3-to-1 speed reduction (and torque increase) on each axis, matching the original reference design's own ratio exactly.

Chapter 3 — Sizing and engineering notes

Gear numbers (verified from the actual 3D model, not estimated)

Module (tooth size)	0.6mm — bumped up from a literal 1/3 scale (0.375mm), which is too fine to 3D-print reliably
Tooth counts	20 : 60 on both axes — unchanged from the original, so the gear ratio is identical
Centre distance	24.0mm on both axes (vs. 15.0mm at a literal 1/3 scale)

ENGINEERING NOTE — why this ended up bigger than the plan's 82x99x90mm target

The plan's ~82x99x90mm figure was a straight one-third linear shrink of the original. Once the real 28BYJ-48 motor's actual dimensions (28mm can, 35mm mounting-tab spacing, 7mm off-centre shaft) went into the model, the final assembly measures roughly **95 x 74 x 96mm**. The tilt motor in particular needs a straight ~22mm run to reach its gear without hitting the tilt shaft's own bearings, which pushed the width out to about 95mm. This is the same kind of gap between an early paper estimate and the real, fully-detailed model that showed up on the sibling redesigns of this same reference file — the motor and bearings are real, fixed-size hardware that can't shrink below their actual dimensions, no matter what the linear-scale math says. It is a sizing finding, not a mistake to silently paper over.

INTERFERENCE CHECK — run with real boolean geometry, not a visual check

Every pair of parts with overlapping bounding boxes (59 candidate pairs) was checked with a real solid-intersection boolean, not eyeballed. Result: all 22 pieces are individually valid solids (no open shells), and the handful of remaining small overlaps are all under 35mm³ — in the same category as the normal, expected contact where two gear teeth interleave at a mesh point (the two largest of these, 10mm³ each, are exactly that). The next-largest handful (20-35mm³) are tight-but-real clearance around the pan motor can and its neighbouring bearing/shaft/spacer stack — worth a small clearance bump in a second revision before cutting real plastic, but not a geometry error.

Assembly steps

1. Press the pan bearing into the main housing's bottom pocket.
2. Slide the main shaft down through the bearing from inside the housing.
3. Thread the lower spacer, then the pan gear (60T, narrow), then the upper spacer onto the main shaft.
4. Bolt the pan motor bracket into the housing, then bolt the pan motor (with its 20T pinion already on its shaft) onto the bracket so the pinion meshes the pan gear.
5. Key Gear 2 Base onto the top of the main shaft.
6. Press the two tilt bearings into Gear 2 Base's bosses; slide Shaft 2 through them.
7. Build up the tilt shaft stack in order: tilt gear (60T, wide), Baring 2 shaft hub, the two small spacers, both tilt bearings.
8. Bolt the tilt motor (with its own 20T pinion) onto Gear 2 Base so its pinion meshes the tilt gear.
9. Rest the Cam Holder's two pivot pins into the sockets on the main housing's twin towers — this joint is not fastened, it just sits there and swings by hand.
10. Clip the laser module into the Cam Holder's front mount.

This guide and the accompanying OPTION_E.step file describe a layout-study-level design: real part counts, real gear ratios, real off-the-shelf hardware sizes, and a verified boolean interference check across the assembled mechanism — not a manufacturing-ready release drawing. Wall thicknesses, fastener callouts, and print tolerances should get a second pass before cutting real plastic.